

BVI-Net: An Ultra-Lightweight Bio-Visually Inspired Network for Skin Lesion Segmentation

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Introduction, Motivation & Problem Statement

Medical Image Segmentation

- Helps clinicians localise lesions in dermoscopy, CT and MRI images.
- Accuracy pursued by scaling models up: U-Net needs 13.4M parameters, VM-UNet 44.3M — too heavy for real clinical hardware.

Problem

- Blurred lesion boundaries and varying lesion sizes make segmentation hard.
- Large models overfit on limited medical data.

Task

Build an **ultra-lightweight** segmentation network, accurate on **skin lesion (ISIC2018)** images.

BVI-Net imitates the biological visual system: a fast **dorsal** stream for global context and a detail-sensitive **ventral** stream for local features, with the global pathway guiding the local one towards the lesion.

- **Local Pathway:** Gabor-initialised, orientation-selective convolution kernels, weights shared across four orientations.
- **Global Pathway (FA-VSSM):** pooling attention gating a state space model, scanned four ways for global context.
- **GCN Attention skips:** features projected into a graph to link non-adjacent regions during upsampling.

Reported: 0.026M parameters | 0.025 GFLOPs | 292 FPS

What We Did

Reimplemented BVI-Net from scratch in PyTorch (no official code released), trained and evaluated on ISIC2018, and compared directly against the paper's reported numbers.

Lightweight models already exist, but each trades away something:

- MALUNet (0.178M), EGE-UNet (0.053M), UltraLight VM-UNet (0.049M) cut parameters sharply.
- CNN-based lightweight models fuse global context poorly.
- Mamba/SSM-based models miss local detail on small lesions.

Gap Identified

No existing model balances global context and local detail at an ultra-lightweight scale. BVI-Net claims to, at 0.026M parameters, but *publishes no code* — leaving the claim unverified.

Objectives and Scope

Objectives

- ① Reimplement BVI-Net from its published description.
- ② Reproduce its reported performance on ISIC2018.
- ③ Verify the claimed 0.026M parameter budget.

Scope

- **Dataset:** ISIC2018 — 2594 dermoscopy images.
- **Task:** Binary skin lesion segmentation.
- **Excluded:** LiTS17, BraTS19, SOTA baseline comparisons.

Dataset

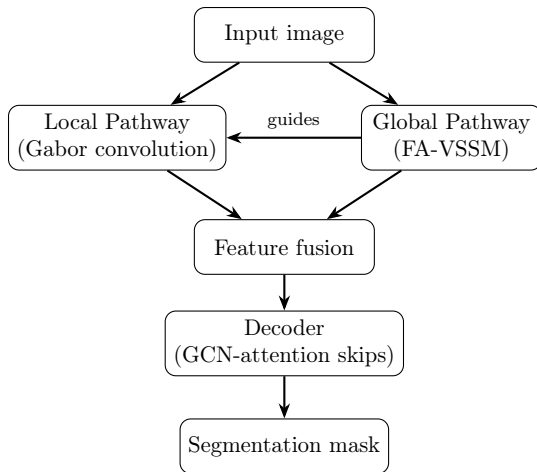
ISIC2018 Task 1 — dermoscopy images with expert-annotated lesion masks. Validation uses ISIC's official 100-image set.

Split	Train	Validation	Test
This work	2335	100	259
Base paper	2594	100	1000

Preprocessing

- Images resized to 256×256 .
- **Training only:** flips, rotations, shift-scale-rotate, brightness/contrast/HSV jitter.
- **Validation & test:** no augmentation.

BVI-Net Architecture



Encoder repeated over 5 stages; the Global Pathway guides the Local Pathway to the lesion region.

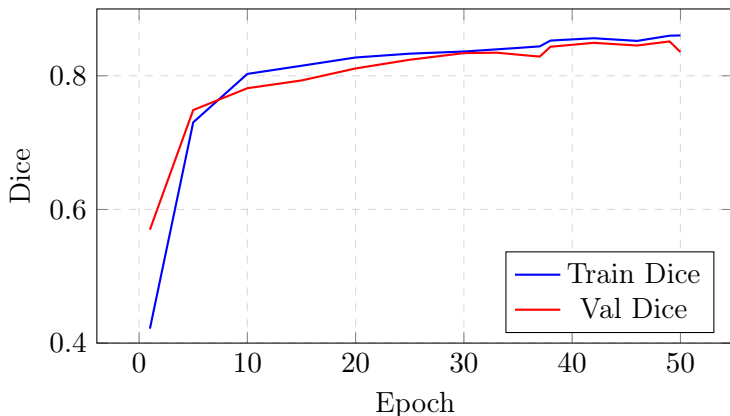
Implementation and Training

Implementation

BVI-Net reimplemented in PyTorch from the paper's textual description. Trained on Kaggle (T4 GPU) using the real `mamba-ssm` selective-scan backend.

Setting	Value
Loss	BCE + Dice
Optimizer	AdamW
Learning Rate	0.001
Batch Size	8
Epochs	50
Early Stopping	Patience = 15
LR Scheduler	$\times 0.5$ on validation-Dice plateau

Training Curve



LR drops push validation Dice to new bests; train/val curves stay close throughout, with no overfitting.

Parameter Reduction Journey

Stage	Params	Val Dice
Initial reconstruction	0.660M	0.8514
+ Width search	0.037M	0.8622
+ True Gabor weight sharing	0.0273M	0.8631
Base paper (claimed)	0.026M	—

- Refactored Gabor convolution into a genuinely shared depthwise kernel + pointwise projection — matching the paper's stated mechanism, not just its initialisation.
- Result: parameter count within **5% of the paper's claim**, with **no loss in Dice**.

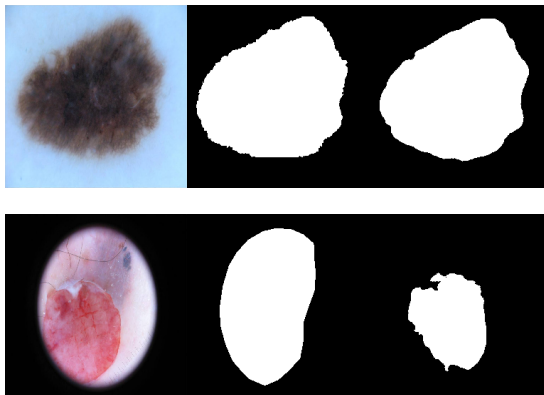
Results: Ours vs. Base Paper

Metric	Base Paper	Ours	Gap
Dice	90.43	86.31	−4.12
mIoU	82.45	78.55	−3.90
Accuracy	95.86	95.44	−0.42
Specificity	97.15	97.80	+0.65
Sensitivity	91.14	88.17	−2.97
Params (M)	0.026	0.0273	+5%

Table 1: Skin lesion segmentation, ISIC2018 (all values in %)

- Parameter count matches the paper’s claim within 5%.
- Specificity exceeds the paper; Accuracy trails by under half a point.
- Dice / mIoU / Sensitivity trail by 3–4 points — a consistent, explainable gap.

Qualitative Results



Left to right: input image, ground truth, prediction.

Top: well-defined lesion — prediction closely tracks the ground truth boundary.

Bottom: prediction under-segments a lesion with a bright, low-contrast surrounding halo, capturing only the darker core region.

Conclusion

- BVI-Net reimplemented successfully with the real Mamba selective-scan backend and true Gabor weight sharing; reaches **86.31 Dice**.
- Specificity **exceeds** the paper (97.80 vs. 97.15); Accuracy trails by only 0.42 points.
- Parameter count (**0.0273M**) matches the claimed **0.026M** within 5% — the ultra-lightweight claim is **verified**.

Key Takeaway

The dual-pathway, bio-inspired design is **reproducible, effective, and genuinely ultra-lightweight** once the paper's weight-sharing mechanism is implemented faithfully.

- Extend replication to **LiTS17** (liver) and **BraTS19** (brain) tasks.
- Ablation study: isolate contribution of Gabor weight sharing, FA-VSSM, and GCN-Attention skips individually.
- Build a **frontend web app** to upload a skin image and view the predicted segmentation in real time.



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Thank you!